

Trackman Progress Report: November 2024 to September 2026

Prepared from 27 range sessions (744 shots) and 5 Trackman Combines (300 scored shots). All figures use Trackman's normalised conditions (Premium ball, 25 C, sea level), so weather and ball type are removed from the comparison. Units are yards and mph throughout; side and curve are signed, positive right and negative left.

Summary for listening

This section is written to be read aloud. The tables that support it follow later in the document.

Over 22 months of Trackman data, speed has been stable and precision has not improved. Eight-iron club speed moved from about 70 miles per hour in the first three sessions to about 73 in the last three, and ball speed from 90 to 94. Five-iron club speed did not move at all, sitting at 80 miles per hour at both ends. Driver speed sits between 89 and 96 depending on the day. That is enough speed for a single-digit handicap. It is not what is holding the score back.

What is holding the score back is dispersion. Roughly three in ten eight-irons and five in ten five-irons finish more than 15 yards off line. The eight-iron has become slightly tighter over the past year, with the share of wide shots falling from 35 percent in the summer of 2025 to 17 percent in early 2026. The five-iron has not: it was 49 percent wide in the summer of 2025 and 38 percent in early 2026, still the loosest club in the bag.

The five Combines tell the same story in one number each: 65, 60, 67, 55 and 66. Flat. The average of the non-disaster shots was between 64 and 71 in every Combine, and 68 or better in four of the five. The difference between a good Combine and a bad one was four blow-up shots against eleven. The long targets are the weak category, and the main cause is the club: the shot to 180 yards has been 11 to 28 yards short every single time, because it is being hit with a club that carries 155.

Three things stand out for the workout program. First, strike quality on the short irons is worth more than speed. The same eight-iron swing speed produces 136 yards on a centred strike and 108 on a poor one. Second, precision decays in the second half of every long session while speed holds, which points at trunk endurance and balance rather than power. Third, the long irons have shallowed out and the miss has flipped from a fade to a hook, a pattern consistent with the lower body stopping and the hands taking over. That is what the hip mobility, pressure-shift and rotation work in the program is for, and the five-iron attack angle is the number that should move if it works.

The practical program for the next eight weeks: train trunk endurance and single-leg stability as hard as strength; hit fewer balls per club and more wedges and drivers; carry a longer club for the 180-yard shot; and score every Combine by counting the shots under 30, not just the total.

Core claim

Speed is not the constraint on this golf game. Over 22 months club speed has been stable within a narrow band (8-iron 69 to 80 mph, 5-iron 76 to 83, driver 89 to 96) and the Combine score has been flat around the low 60s. The gap between the current game and a mid-single-digit handicap is dispersion: roughly a third of 8-iron shots and half of 5-iron shots finish more than 15 yards off line, and every Combine has been decided by the number of blow-up shots, not by the quality of the average shot.

The practical implication is that the physical program should be judged on repeatability first (strike location, face control late in a session, long-iron attack angle) and on speed second.

What the data covers

	Sessions	Shots	Period
8-iron	23	415	Nov 2024 to Jan 2026
5-iron	18	219	Jan 2025 to Jan 2026
Driver	8	29	Feb 2025 to Sep 2026
3-wood	5	33	Mar 2025 to Oct 2025
7-iron	2	20	Sep 2026
Wedges (PW, 58, lob)	3	22	May to Sep 2025
Combines	5	300	Apr 2025 to Aug 2026

Two things about the data shape the interpretation. The Trackman export format changed twice: sessions before August 2025 report curve, club path and face-to-path but not side (finishing position), and sessions after report side but not path. Dispersion statistics below therefore come from August 2025 onward, and path statistics mostly from before. Second, the driver and wedges are barely represented; 56 percent of all recorded shots are 8-irons. That is a finding in itself (see Finding 7).

Progress over time

Pooling every shot by calendar quarter removes the session-to-session noise and shows what has actually moved. Smash factor is missing for 2025 Q4 because the export layout in that period did not include it.

8-iron by quarter

Quarter	Sessions	Shots	Club speed	Ball speed	Smash	Carry	Carry SD	Side SD	Beyond 15 yds	Attack	Spin
2024 Q4	2	79	69.9	89.5	1.28	114	10.4	n/a	n/a	-1.4	6,200
2025 Q1	2	22	72.5	94.4	1.30	124	6.9	n/a	n/a	-2.9	6,090
2025 Q2	2	25	76.1	99.9	1.31	129	26.8	n/a	n/a	-3.2	5,370
2025 Q3	6	101	74.3	98.1	1.31	131	13.8	15.0	35%	-2.4	5,760
2025 Q4	8	125	74.7	96.2	n/a	123	30.1	12.9	26%	-3.1	5,920
2026 Q1	3	63	72.6	93.7	1.31	121	14.8	11.7	17%	-2.0	5,860

Reading: speed rose about 4 mph from late 2024 to mid 2025 and has given half of it back. Carry followed. The dispersion trend is the encouraging one: side SD has tightened every quarter it has been measured (15.0, 12.9, 11.7 yards) and the share of shots beyond 15 yards halved (35 to 17 percent). Carry SD has not improved; the 30-yard figure in 2025 Q4 is driven by a handful of fat and thin shots in two sessions, which is the strike-quality problem in Finding 1. The share of well-struck 8-irons (smash 1.35 or better) was 15 percent in late 2024 and 25 percent in early 2026, a gain, but three in four shots are still below that mark.

5-iron by quarter

Quarter	Sessions	Shots	Club speed	Ball speed	Smash	Carry	Carry SD	Side SD	Beyond 15 yds	Attack	Spin
2025 Q1	3	60	80.0	108.0	1.35	152	13.1	n/a	n/a	-1.7	4,810
2025 Q2	2	15	78.6	111.1	1.41	156	14.5	n/a	n/a	-4.0	4,220
2025 Q3	6	102	81.1	111.2	1.37	157	21.3	21.7	49%	-0.9	4,220
2025 Q4	4	21	78.8	108.7	n/a	151	16.9	17.2	48%	-1.2	4,220
2026 Q1	3	21	80.4	110.7	1.41	158	16.8	17.0	38%	-0.8	4,090

Reading: club speed has not moved (80.0 to 80.4), but the strike has improved: smash from 1.35 to 1.41 and ball speed from 108 to 111 on the same club speed, worth six yards of carry. The cost has been the attack angle, which went from -1.7 to -0.8 and spent the summer of 2025 near zero, and with it spin fell 700 rpm. Lateral dispersion improved from the summer (21.7 to 17.0 yards SD) but 38 percent of 5-irons still finish more than 15 yards wide. This club has traded control for a slightly better strike, which is not a good trade at 160 yards.

Driver and 3-wood

Too few shots for quarterly tables (29 and 33 shots respectively). Driver club speed across the multi-shot sessions: 94.2 (Feb 2025), 93.6 (Q3 2025), 91.2 (Q4 2025), 88.9 (Sep 2026, four shots). Ball speed 135, 132, 125, 133. Smash has improved from 1.44 to 1.48 while speed has drifted down about 4 mph, so carry has held around 190 to 210. The direction of the speed drift is worth watching given the shoulder rebuild; the 10-ball driver test is the right instrument.

First three sessions against the last three

Club	Metric	First three	Last three	Change
8-iron (Nov 2024 to Feb 2025 vs Jan 2026)	Club speed	70.5	72.6	+2.1 mph
	Ball speed	90.2	93.7	+3.5 mph
	Smash	1.28	1.31	+0.03
	Carry	115.6	120.8	+5.2 yds
	Carry SD	10.5	14.8	+4.3 yds (worse)
5-iron (Jan to Mar 2025 vs Jan 2026)	Club speed	80.0	80.4	+0.4 mph
	Ball speed	108.1	110.8	+2.7 mph
	Smash	1.35	1.41	+0.06
	Carry	151.6	157.7	+6.1 yds
	Carry SD	13.1	16.8	+3.7 yds (worse)
	Attack angle	-1.7	-0.8	+0.9 deg (shallower)

Both irons gained five or six yards of carry over the year, almost entirely from strike efficiency rather than speed, and both got looser on distance control. Longer but less predictable is the summary of the year.

Combine trajectory

Combine	Score	Est. hcp	Shots below 30	Mean of shots 30 and above	Shots 80 and above	Driver ball speed	Driver side SD
2025-04-02	65.2	8	5	70.6	17	133.6	21.0
2025-05-13	59.7	11	8	68.0	13	124.3	10.0
2025-06-10	66.5	7	4	70.2	19	134.0	7.7
2025-07-24	54.6	16	11	64.4	12	140.9	11.8
2026-08-25	65.6	8	5	70.5	20	129.8	7.3

The column that moves the total is the blow-up count. The mean of the good shots sits at 68 to 71 in four of five tests, and the number of excellent shots (80 or better) was highest in the most recent Combine. The driver has tightened markedly (side SD 21 yards in the first Combine, 7 in the last) at the cost of about 4 mph of ball speed, which is a reasonable trade during a shoulder rebuild and matches the pattern in the range data.

Findings

1. Speed is stable. Distance variation is strike quality.

Session-average 8-iron club speed ranged from 68.7 to 79.9 mph with a slight upward drift (about +2.6 mph per year on a regression of session means). 5-iron club speed is flat at 80 mph. The three multi-shot driver sessions sit at 92 to 96 mph. None of this is moving fast enough to explain the session-to-session swings in carry, which for the 8-iron ran from 99 to 145 yards.

Smash factor explains those swings. Pooling every 8-iron shot with club data, shots struck at a smash factor of 1.35 or better carried 136 yards at 72.9 mph of club speed; shots below 1.25 carried 107.5 yards at 71.5 mph. Same speed, 28 yards apart. Only 21 percent of 8-iron shots reached 1.35, and 19 percent fell below 1.25. The 5-iron and driver are better (1.37 and 1.43 on average), which is unusual: most golfers strike short irons more efficiently than long clubs, so the 8-iron figure points at a delivery issue with the scoring clubs rather than a general speed problem.

Implication: a centred strike is worth more than any realistic speed gain. Adding 3 mph of 8-iron club speed is worth roughly 7 yards; moving the median strike from 1.30 to 1.36 is worth roughly 10.

2. Lateral dispersion is the scoring problem

From August 2025 onward (the sessions that report finishing position):

Club	Shots	Side SD	Beyond 15 yds off line	Bias
8-iron	230	13.8 yds	29%	2.6 yds right
5-iron	124	20.8 yds	48%	1.8 yds left
7-iron (2026)	14	9.1 yds	21%	7.8 yds right
Driver	11	13.8 yds	45%	9.1 yds left
3-wood	13	21.4 yds	54%	1.5 yds left

A 130-yard 8-iron with a 14-yard side SD means the one-standard-deviation band is 28 yards wide. That is the width of a green. The 5-iron's band is 42 yards. Carry control is similarly loose: 8-iron carry SD is about 10 yards on a good day and 15 to 25 on others, with individual sessions containing 30-to-100-yard mishits.

The bias numbers are small on average but hide a two-way miss. The 8-iron and 7-iron leak right: in ten of the fifteen 8-iron sessions with side data, at least 63 percent of shots finished right of the line. The 5-iron and driver miss left, and the 5-iron's left misses in late 2025 were large: session averages of 17, 21 and 16 yards left in December and January.

3. The miss pattern has flipped since 2024

The path and face data tell the story behind the two-way miss.

Period	Club	Path	Face to path	Pattern
Dec 2024	8-iron	-8.9 (out-to-in)	+5.0 (open)	Pull-fade, weak right
Mar 2025	5-iron	-5.6	+4.5	Pull-fade
Feb 2025	Driver	+7.5 (in-to-out)	-1.1 (closed)	Draw, finishing left
Jul 2025	Driver	+5.8	-4.9	Hook
Jan 2026	8-iron	+3.6	-2.1	Draw
Jan 2026	5-iron	+4.9	-4.1	Hook
Sep 2026	7-iron	-5.3	+1.0	Pull-fade

In 2024 the irons were out-to-in with an open face. Through 2025 the path swung to in-to-out and the face closed, first with the driver and then with the irons. By January 2026 the 8-iron was 3.6 degrees in-to-out with a closed face, and the 5-iron 4.9 degrees. The 7-iron in September 2026 is back to out-to-in.

Two readings are possible. One is that a swing change over-corrected the slice into a hook and is now being corrected back. The other, which the numbers support more directly, is that path is

repeatable while face is not: path SD within a session is typically 1 to 3 degrees, face-to-path SD 2 to 5.5 degrees and almost always the larger of the two. The dispersion problem is a face-control problem. For the driver specifically, every session since February 2025 that measured path shows in-to-out with a closing face, and the average finish was left in three of the five sessions with side data. The workout program's assumption of a right-miss, slice-pattern driver is out of date.

4. The long irons are shallowing out

5-iron attack angle drifted from -4.6 degrees (April 2025) to +0.7 and +1.2 (December 2025 and January 2026), a change of about +2 degrees per year. Hitting up on a 5-iron is rare and it costs control. Splitting all 5-iron shots by attack angle:

Attack angle	Shots	Spin	Carry SD	Side SD
Zero or positive	47	3,600 rpm	20.9 yds	20.3 yds
Steeper than -2	66	4,900 rpm	15.5 yds	18.4 yds

The shallow shots spin 1,300 rpm less, which is why the 5-iron carry range in a single session can span 50 yards, and they coincide with the in-to-out, closed-face, left-miss sessions. A shallow attack with an in-to-out path and a closing face is the signature of a lower body that stops rotating so the arms and hands take over to square the club. That is a hypothesis to test with a coach, not a conclusion from launch data alone, but it is the pattern the program's hip mobility and pressure-shift work is designed to address, and it gives those exercises a measurable target.

The driver is the opposite and healthy: attack angle +2 to +4 degrees, spin 2,100 to 2,900 rpm, smash 1.43 to 1.48. The two low-spin outliers (1,660 and 1,992 rpm) came with the highest attack angles and shorter carries, which is the ceiling on hitting up.

5. The Combine score is a count of blow-ups

Five Combines: 65.2, 59.7, 66.5, 54.6, 65.6 (estimated handicaps 8, 11, 7, 16, 8). Flat over 16 months.

Removing every shot scored below 30, the five Combines would read 70.6, 68.0, 70.2, 64.4 and 70.5. The best Combine had 4 blow-ups; the worst had 11. Everything separating a 66 from a 55 lives in those shots. The median shot was 66 to 72 in four of the five Combines; the typical shot has not changed, and the average is set by how many disasters happen.

By distance category:

Category	Apr 2025	May 2025	Jun 2025	Jul 2025	Aug 2026
60 to 100 yds	67	60	66	52	71
120 to 140 yds	75	64	64	53	74
160 to 180 yds	54	56	67	59	44
Driver	57	60	74	61	66

The wedge and mid-iron categories improved in the most recent test. The long category got worse, and it has a specific cause: distance, not direction. The average shot at the 180-yard target carried 152, 160, 169, 167 and 159 yards, short by 11 to 28 yards every time. The 160-yard target was short by 7 to 10 yards in four of five tests. The club being used for 180 (club speed 80 to 85 mph, which matches the 5-iron) carries 155 yards. The 180 shot needs a 4-iron, hybrid or 3-wood, and the Combine has been penalising a club-selection decision for over a year.

6. Speed holds up within a session. Precision does not.

Comparing the first half of a session to the second half for every club with at least 14 shots: club speed is flat or slightly higher late (the 50-shot 8-iron session in November 2024 went from 67 to 70 mph, a warm-up effect), while dispersion widens. 5-iron carry SD went from 20 to 25 and 16 to 21 yards in the two long August 2025 sessions; the September 2026 7-iron session went from a carry SD of 3.9 to 9.0 and a side SD of 7.3 to 10.2 over 14 shots.

In practice this means the body is not running out of speed; it is running out of repeatability, and the second half of a 15-to-20-ball block with one club is measurably looser than the first. The program's trunk-endurance and balance work (side plank, carries, single-leg balance) maps to this directly, and it also argues for structuring range sessions as short blocks that alternate clubs rather than 30 to 50 balls with one iron.

7. Practice time is not going where the Combine loses points

Club group	Share of practice shots	Combine category performance
8-iron	56%	120 to 140 yds: strongest category
5-iron	29%	160 to 180 yds: weakest, and the wrong club for 180
Driver	4%	57 to 74, two-way miss
Wedges	3%	60 to 100 yds: 52 to 71, most blow-ups by count

The 60 to 100 yard targets produced 16 of the 33 blow-up shots across all five Combines (the 60-yard target alone produced 7), and the wedges have 22 recorded practice shots in 22 months. The 8-iron, which receives the majority of practice, is the club the Combine rewards most. The allocation is inverted relative to where the score is lost.

Implications

For the physical program

The program's north star is "a faster, more repeatable swing." The data say to weight the second half. Suggested emphasis by training day, with the Trackman metric that should move if the work transfers:

Program element	Data pattern it addresses	Metric to watch
Day 2 and Day 5 strength (goblet squat, RDL, split squat, deadlift)	Speed plateau at 73 / 80 / 93 mph	10-ball 7-iron and driver ball speed at controlled effort
Day 3 trunk endurance (side plank, dead bug, carries)	Second-half dispersion widening	Side SD and carry SD, shots 1 to 8 vs 9 to 16
Day 4 rotation and pressure shift (lateral skater step, band rotation, golf-posture rotation) and 90/90 hip work	In-to-out path, closing face, shallow 5-iron attack angle	5-iron attack angle back below -2; face-to-path SD below 3 degrees
Single-leg balance, split-squat symmetry	8-iron smash spread (19% below 1.25)	Share of 8-iron shots at smash 1.35 or better

Two cautions. The driver assumption in the program should be rewritten from "right miss/slice" to "two-way miss, left on average, with an in-to-out path." And the physical benchmarks (90/90 control, side-plank symmetry) should be logged with left/right values; if there is a lead-hip internal rotation asymmetry, it is the most likely physical link to the pattern in Finding 4.

Workout pointers for the next cycle

These stay inside the program's existing exercise catalogue and safety rules (soreness above 3 out of 10 means Recovery Flow, nothing the clinician has not cleared for the shoulder). The point is to weight the work toward what the Trackman data say is limiting the score.

1. **Treat trunk endurance as the primary quality, not a warm-up.** The data show speed holding and precision decaying within a session. Side plank, dead bug, bird dog and suitcase carries are the Day 3 block; give them the same progression discipline as the lifts. Target a clean 40-second side plank each side with less than 10 percent left-right difference, and suitcase carries at 40 seconds per side without a lean. Retest: second-half side SD in a 16-ball 7-iron block should close toward the first half.
1. **Own the lead hip.** The in-to-out path, closing face and shallow 5-iron attack angle are the signature of a pelvis that stops turning through impact. Daily 90/90 transitions (progress to no hands, then lift-offs), adductor rock-backs and couch stretch build the range; lateral skater steps and step-ups teach loading and unloading the lead side. Log left and right separately. If the lead (left, for a right-handed golfer) hip has less internal rotation, that asymmetry is the first thing to close. Retest: 5-iron attack angle back below -2 degrees.
1. **Single-leg stability for strike location.** Nineteen percent of 8-irons are struck poorly enough to lose 20-plus yards. Single-leg balance (progress to eyes closed, arms crossed) and split squats with a strict 3-1-1-0 tempo train the ability to keep the low point where it was at address. Retest: share of 8-iron shots at smash 1.35 or better, currently 25 percent.

1. **Strength for speed, but measured, not chased.** Goblet squat, RDL, trap-bar deadlift and rear-foot-elevated split squat at RPE 6 to 7, two to three good reps in reserve, 2 to 5 percent load increases only after clean sessions. The 8-iron has picked up 2 mph over the year while the 5-iron has not, so a realistic eight-week target is +2 mph of 7-iron club speed at the same effort with no loss in side SD. Anything faster than that is likely to come from swinging harder, which the dispersion data say will cost more than it gains.
1. **Rotational power only after the pattern is stable.** Squat-to-quick-stand, band rotation with the ground-hips-torso-arms sequence, and golf-posture rotations belong on Day 4, but keep the intent on sequencing, not load. A medicine-ball throw progression can be added later if the shoulder is cleared; it is not in the current catalogue and should not be improvised.
1. **Warm up like it matters, because the data say it does.** The first half of long sessions is 3 mph slower than the second, and the wide first shots in Combines are where the blow-ups start. The program's 10 to 15 minute golf warm-up (walk, cat-cow, 90/90, open book, posture rotations, progressive swings at 50 to 80 percent) should precede every range session, every test and every round.
1. **Protect the shoulder budget.** Driver ball speed has drifted down about 4 mph while driver dispersion has tightened. That is an acceptable trade during a rebuild. Do not spend shoulder tolerance on maximum-speed driver work; spend it on the PT block and on pulling strength (supported rows without shrugging), and let speed return through the legs and hips.
1. **Measure on a fixed cadence.** Physical benchmarks every 2 to 4 weeks with left and right values; 10-ball 7-iron and 10-ball driver tests every 2 to 4 weeks, exported as their own Trackman PDF; a full Combine every 4 to 8 weeks on a rested day. Six weeks from now the question is simple: did 5-iron attack angle steepen, did second-half side SD close, did the smash 1.35 share rise, and did the blow-up count fall. If the gym numbers improve and those four do not, change the transfer work before adding load.

For practice

1. Make face control the technical priority. Path is already repeatable; face-to-path variance is the source of the two-way miss. Any drill that gives face feedback at impact (alignment stick gate, impact tape, a coach's face-angle read) is the right kind.
2. Fix the 180-yard club. Carry the 4-iron, a hybrid or the 3-wood on the Combine's 180 target and hit that club in practice. This is the single cheapest Combine improvement available.
3. Restore a downward strike with the 5-iron. Ball position and weight forward at setup are the usual first checks; the target is an attack angle of -2 to -4 with spin back above 4,500 rpm.
4. Rebalance practice toward 60 to 100 yard wedges and the driver. Move to roughly one Combine-style session in three: random targets, six balls each, scored.

5. Cap block practice at 8 to 10 balls per club and rotate. Dispersion widens after that point, so a 40-ball 8-iron block trains the loose pattern as much as the good one.

For the mental side

The Combine result is a blow-up count, so the mental target is not "hit better shots" but "eliminate the shot below 30." That reframes course management. On long targets the miss that costs the most is the one that is both short and wide; a conservative club that carries the number and a target that accepts a 10-yard miss would have scored better in every Combine. A written pre-shot routine and a decision to accept the boring shot on the 160 to 180 targets is worth more than a swing thought.

The warm-up matters more than it might feel. The first half of long sessions is 3 mph slower and the Combine's worst category performances came with wide first shots. Use the program's 10 to 15 minute golf warm-up before every test and before the first tee.

Finally, use the Combine as the KPI and score it three ways: total, mean of shots scored 30 and above, and count of shots below 30. The first tells you the result, the other two tell you which lever moved.

Baseline values for the program scorecard

Filling in the "Baseline" column of the program's eight-metric scorecard from the most recent data:

Metric	Baseline	Source
Combine score	65.6	2026-08-25
Combine shots below 30	5 of 60	2026-08-25
7-iron carry (controlled effort)	130 yds	Sep 2026, 20 shots
7-iron side SD	9.1 yds	Sep 2026, 14 shots with side data
7-iron smash factor	1.31	Sep 2026
Driver ball speed	132.7 mph	Sep 2026, 4 shots
Driver side SD	23.6 yds	Sep 2026, 2 shots with side data
5-iron attack angle	-1.2 deg (Jan 2026), target below -2	Jan 2026

The driver baseline needs a proper 10-ball test; 4 shots is not a baseline.

Data notes

- The PDFs have no text layer. Every number was OCR'd and cross-checked against the Average row Trackman prints on each report; all 27 sessions reconcile on every column.
- Trackman's session average excludes shots with missing data. This report does the same.

- "Curve" (pre-August 2025) and "Side" (post-August 2025) are different measurements. Curve is how much the ball bent; side is where it finished. They are not combined anywhere in this report.
- Combine "from pin" is in feet; everything else is in yards.